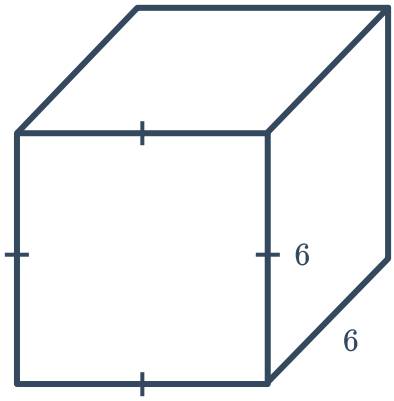
Surface area of prisms

1 Consider the following cube with a side length of 6 cm:

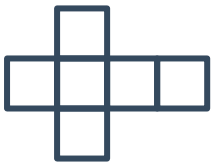


a Which of the following nets match the given cube?

A



B



C



D

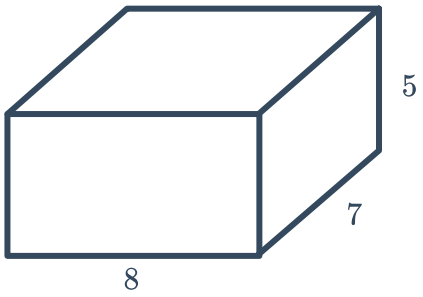


b Find the area of the cube's net.

c Find the surface area of the cube.

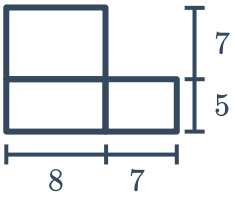
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2 Consider the following rectangular prism:

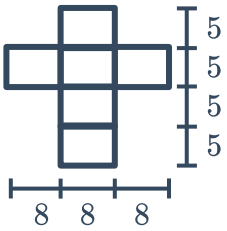


a Which of the following nets match the given rectangular prism?

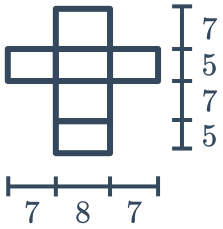
A



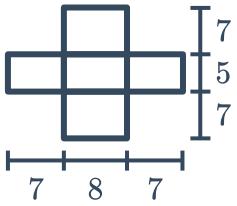
B



C



D

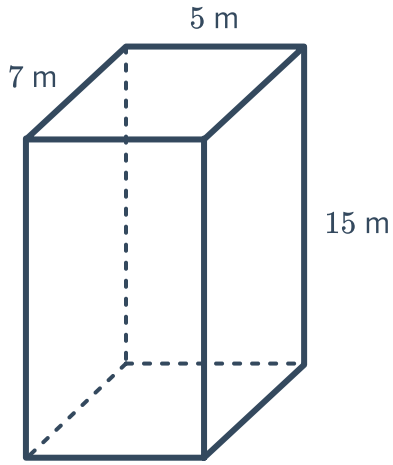


b Find the area of the prism's net.

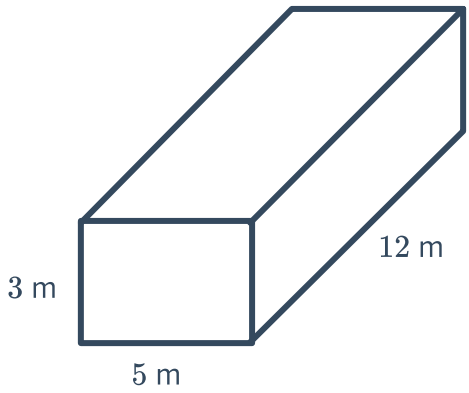
c Find the surface area of the rectangular prism.

3 Find the surface area of the following rectangular prisms:

a

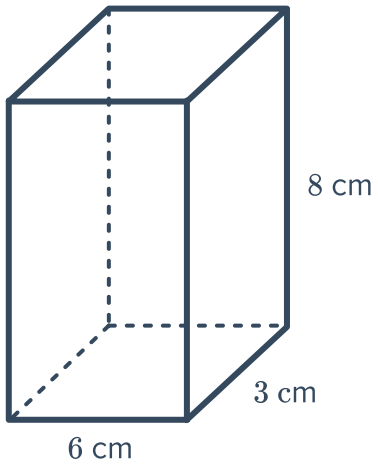


b

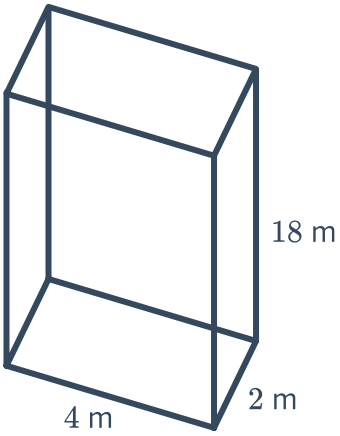


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c

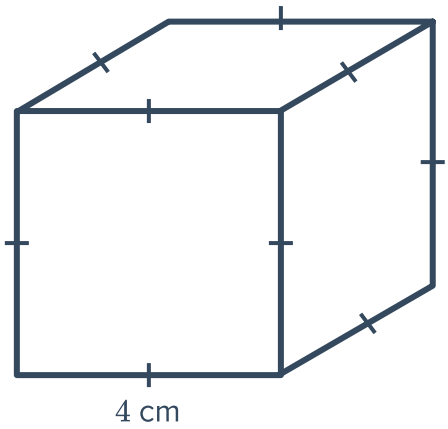


d

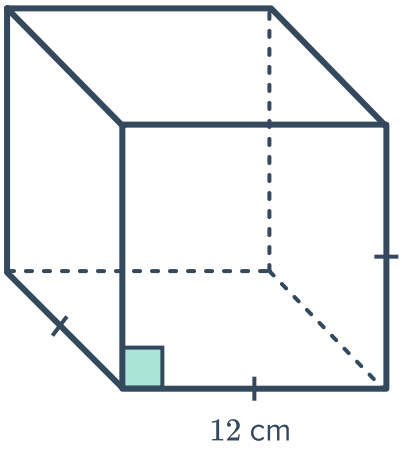


4 Find the surface area of the following cubes:

a



b

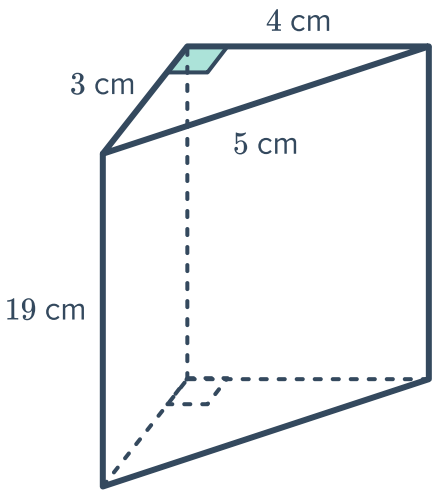


5 Find the side length of a cube that has a surface area of $384\ \text{cm}^2$.

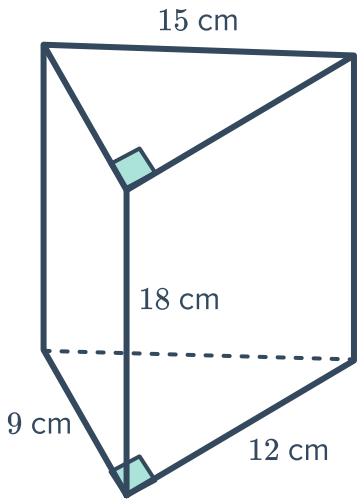
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6 Find the surface area of the following triangular prisms:

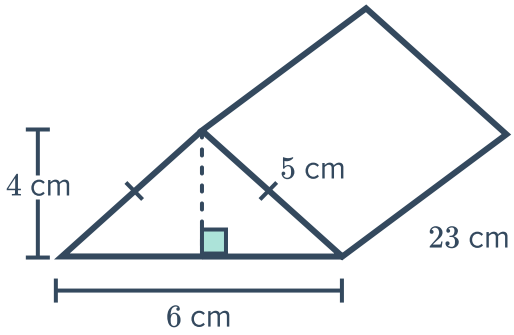
a



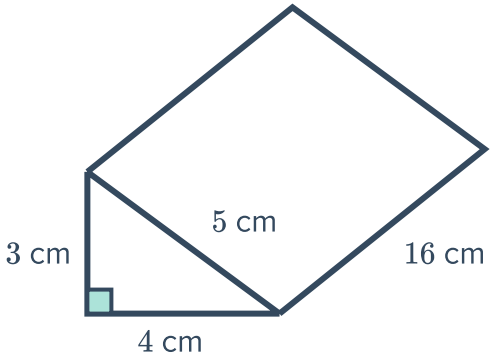
b



c



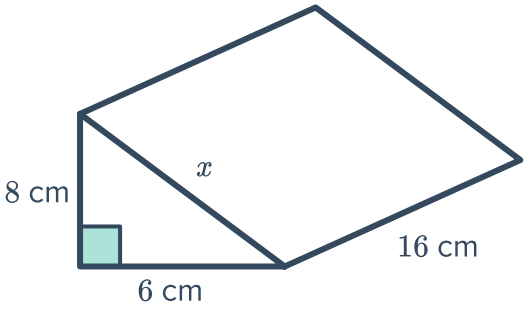
d



7 Consider the following figure:

a Find the value of x.

b Find its surface area.

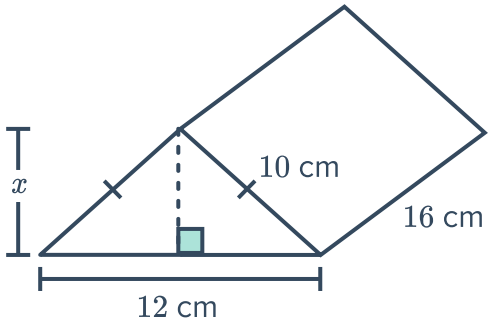


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8 Consider the following figure:

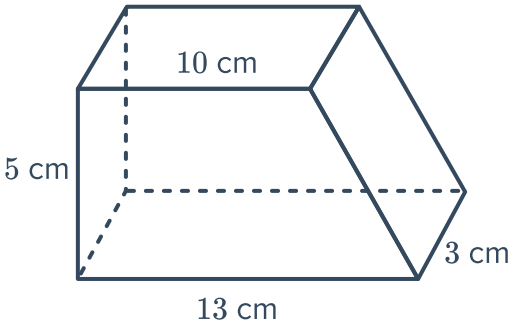
a Find the find the value of x.

b Find its surface area.

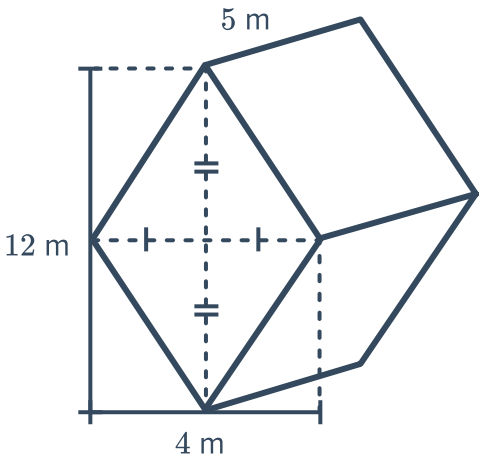


9 Find the surface area of the following prisms. Round your answer to two decimal places when necessary.

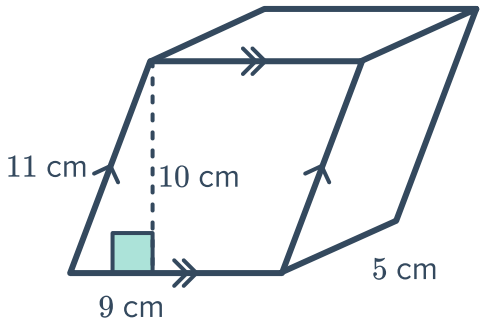
a



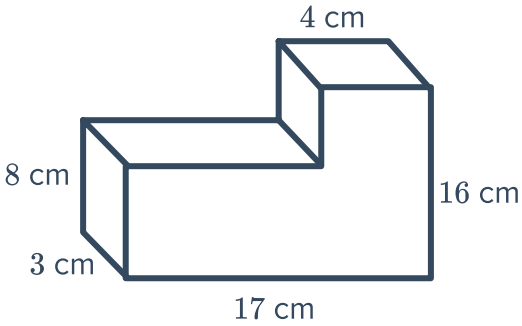
b



c



d

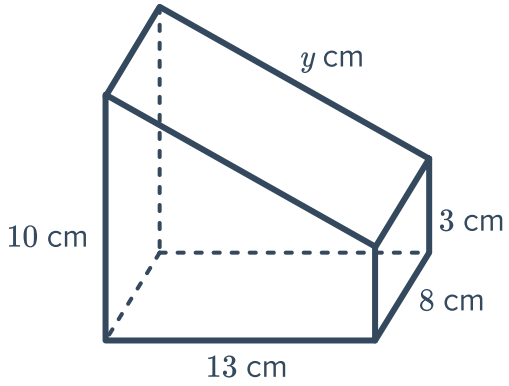


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10 Consider the following trapezoidal prism:

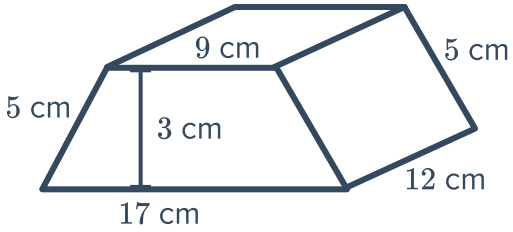
a Find the value of y correct to one decimal place.

b Find its surface area.



11 Consider the following trapezoidal prism:

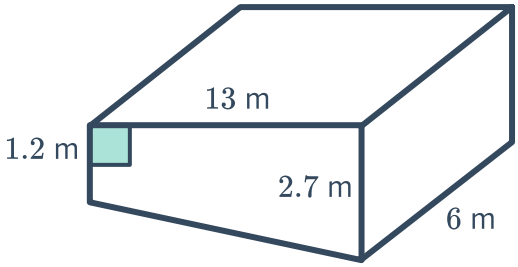
Find its surface area.



Applications

12 Consider the swimming pool with the following dimensions:

Calculate the area inside the pool that is to be tiled. Round your answer to two decimal places.



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13 Sally is building a storage chest in the shape of a rectangular prism. The chest will be 93 cm long, 80 cm deep, and 18 cm high. Find the surface area of the chest.

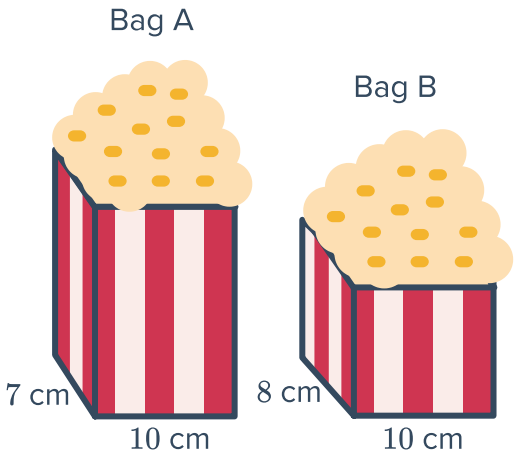
14 A birthday gift is placed inside the box shown:

Find the minimum amount of wrapping paper needed to wrap this gift?



15 Both of the following popcorn bags are designed to carry 735 cm^3 of popcorn.

- **a** Find the height h of bag A.
- **b** Find the height h of bag B.
- **c** Find the surface area of bag A.
- **d** Find the surface area of bag B.
- **e** Which bag should be used to reduce the amount of paper needed?



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